

... of the elements below, is a nonmetal

- A) K (atomic no. = 19)
- B) Mg (atomic no. = 12)
- C) Li (atomic no. = 3)
- D) Cl (atomic no. = 17)
- E) Ge (atomic no. = 32)

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☒ D
- ☐ E

How many electrons are there in the valence shell of the Si⁴⁺ ion?

(atomic number =14)

A. 2

B. 8

C. 7

D. 6

E. 4

Select one:

☐ A

☐ B

☐ C

☒ D

☐ E

The principal quantum number, _____ (n) of the first p- orbital is

A. 1

B. 2

C. 3

D. 4

E. 0

إختتر أحد الخيارات:

A ☐

B ☒

C ☐

D ☐

The molar mass of compound

$$\text{S}_4\text{O}_x = 224 \text{ g/mol}$$

Atomic weights: S = 32, O = 16
g/mol

?What is the value x

A. 3

B. 5

C. 7

D. 4

E. 6

إختتر أحد الخيارات:

A ☐

B ☐

C ☐

**1s 2s 2p 3s 3p 4s 3d is a
ground state configuration for a**

- A. Transition element**
- B. Representative element**
- C. Halogen**
- D. Inner transition element**
- E. Nonmetal element**

إختتر أحد الخيارات:

A ☒

B ☐

C ☐

D ☐

E ☐

**what is the atomic number for
an element that belongs to
,period 5
?group 2B in the periodic table**

A. 47

B. 42

C. 41

D. 80

E. 48

إختر أحد الخيارات:

A ☐

B ☐

C ☐

D ☐

Which element has the lowest electron affinity (EA) for



A. F

B. Li

C. Ne

إختتر أحد الخيارات:

A ☐

B ☐

C ☐

(Atomic weight of Ca = 40 g/mol)

- A. 1.0 mole
- B. 0.5 mole
- C. 2.5 mole
- D. 2.0 mole
- E. 1.5 mole

إختتر أحد الخيارات:

A ☐

B ☐

C ☒

D ☐

E ☐

The element X (atomic number = 30) is a _____ element with unpaired electrons _____

A. paramagnetic , 3

B. paramagnetic , 2

C. paramagnetic, 6

D. diamagnetic , 2

☒ E. diamagnetic , 0

إختتر أحد الخيارات:

A ☐

B ☐

**Which of the elements below, is
a nonmetal**

- A) K (atomic no. = 19)**
- B) Mg (atomic no. = 12)**
- C) Li (atomic no. = 3)**
- D) Si (atomic no. = 14)**
- E) S (atomic no. = 16)**

إختَر أحد الخيارات:

A ☐

B ☐

C ☐

D ☐

E ☒

of atoms) are found
in 7.5 mole of $K_2Cr_2O_7$

Avogadro's number =
 6.022×10^{23}

A. 2.3×10^{25}

B. 7×10^{25}

C. 5×10^{25}

D. 1.4×10^{26}

E. 3.6×10^{24}

إختر أحد الخيارات:

A ☐

The quantum number that describes the energy of an orbital is

- A.** Principle quantum number
- B.** Magnetic quantum number
- C.** Secondary (angular momentum) quantum number
- D.** Spin quantum number
- E.** None of the above

إختتر أحد الخيارات:

A ☐

B ☐

C ☐

D ☐

The first ionization energy for elements _____ from LEFT to RIGHT through a period and _____ from BOTTOM to the TOP through a group in the periodic table.

- A. increase, increase
- B. increase, decrease
- C. decrease, increase
- D. decrease, decrease
- E. is constant, decrease

Select one:

☒ A

☐ B

☐ C

☐ D

☐ E

The quantum number that describes the shape of an orbital is:

- A. Principle quantum number
- B. Magnetic quantum number
- C. Secondary (angular momentum) quantum number
- D. Spin quantum number
- E. None of the above

Select one:

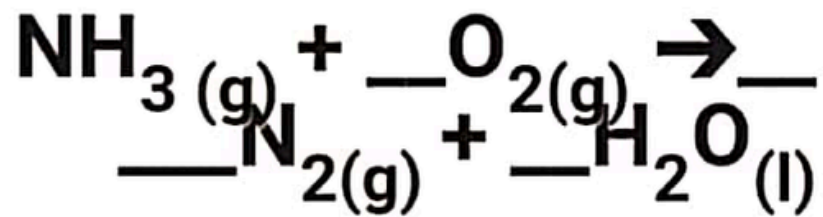
☐ A

☐ B

☒ C

☐ D

☐ E



The coefficients in the balanced equation : (respectively) are

- A. 4,3,2,6
- B. 2,1,2,3
- C. 1,3,1,3
- D. 2,3,2,3
- E. 3,4,6,2

إختار أحد الخيارات:

- A ☐
- B ☐
- C ☐
- D ☐

Calculate the mass percentage of nitrogen (N) in
of $\text{Al}(\text{NO}_3)_2$
(Atomic weights: Al= 27 , N=14, O =16 g/mol)

- A. 17.1%
- B. 19.7%
- C. 58.5%
- D. 67.6%
- E. 12.2%

Select one:

☐ A

☒ B

☐ C

☐ D

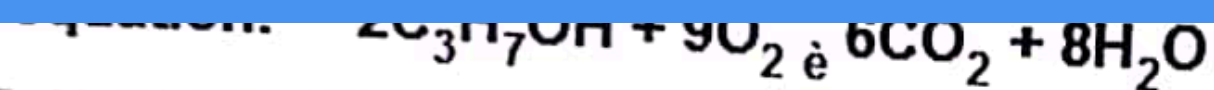
☐ E

the secondary quantum number (l) is:

- A. 1
- B. 3
- C. 4
- D. 0
- E. 2

Select one:

- ☐ A
- ☐ B
- ☒ C
- ☐ D
- ☐ E



If 12 g $\text{C}_3\text{H}_7\text{OH}$ (60 g/mol) are allowed to react with 40 g O_2 (32g/mol)
The mass of CO_2 (44g/mol) produced in grams is:

- A. 13.75
- B. 18.33
- C. 22.91
- D. 26.4
- E. 17.6

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☒ D
- ☐ E

The molar mass of compound $\text{S}_2\text{O}_x = 112 \text{ g/mol}$

Atomic weights: $\text{S} = 32$, $\text{O} = 16 \text{ g/mol}$

What is the value x ?

A. 3

B. 5

C. 7

D. 4

E. 6

Select one:

☒ A

☐ B

☐ C

☐ D

☐ E

How many moles is 100 g of Ca?
(Atomic weight of Ca = 40 g/mol)

- A. 1.0 mole
- B. 0.5 mole
- C. 2.5 mole
- D. 2.0 mole
- E. 1.5 mole

Select one:

☐ A

☐ B

☒ C

☐ D

☐ E

How many atoms (total number of atoms) are found in 0.5 mole of $K_2S_4O_6$

Avogadro's number = 6.022×10^{23}

- A. 2.3×10^{25}
- B. 7×10^{25}
- C. 5×10^{25}
- D. 1.4×10^{26}
- E. 3.6×10^{24}

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☒ E

which element has the lowest IE_1 in period 2 ?

A. Ne

B. Ar

C. Li

D. Na

Select one:

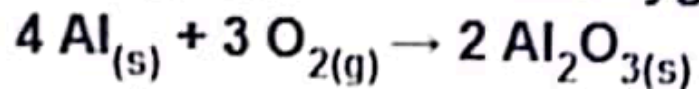
☐ A

☐ B

☒ C

☐ D

Aluminum reacts with oxygen to produce aluminum oxide.



If 10.0 moles of Al react with excess O_2 , how many moles of Al_2O_3 can be formed?

- A. 1.5 mol
- B. 2.0 mol
- C. 2.5 mol
- D. 5.0 mol
- E. 3.0 mol

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☒ D
- ☐ E

...ground state atom has the
quantum number set (n, l, m_l, m_s) equal to $(2, 0, 0, \frac{1}{2})$
(Atomic no. of N = 7)

- A. first electron
- B. fourth electron
- C. fifth electron
- D. sixth electron
- E. seventh electron

Select one

☐ A

☒ B

☐ C

☐ D

☐ E

Which electron in the nitrogen ground state atom has the quantum number set (n, l, m_l, m_s) equal to $(2, 1, 0, \frac{1}{2})$
(Atomic no. of N = 7)

- A. first electron
- B. third electron
- C. fifth electron
- D. sixth electron
- E. seventh electron

Select one:

☐ A

☐ B

☒ C

☐ D

If 0.6 mol ZnS are allowed to react with excess amount of O_2
Calculate percentage yield of the reaction given that 10 g of
 SO_2 (64g/mol) is actually produced :

- A. 86.8 %
- B. 60 %
- C. 43.4%
- D. 26 %
- E. 30 %

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☒ D